Andrew M. C. Wong

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LinkedIn | GitHub | Website

EDUCATION

University of Toronto | *Master of Science in Applied Computing* **Toronto, Ontario**
AI in Healthcare Concentration *September 2026 - Present*
University of Toronto | *Bachelor of Science* | *(cGPA: 3.85 / 4.00)* **Toronto, Ontario**
Bioinformatics and Computer Science *September 2021 - June 2026*

EXPERIENCE

Research Fellow | Agentic AI in Education **Toronto, Ontario**
Ontario Institute for Studies in Education of the University of Toronto *April 2025 - June 2026*

- Secured a $120,000 Mitacs Accelerate grant by initiating a partnership between Factors Education and OISE for research related to the effects of AI on early education.
- Designed and deployed an AI agent orchestration system with 15+ specialized agents and multi-phase automation for K-12 lesson planning.
- Reduced agent instruction costs by ~90% through automated scenario testing and Braintrust observability, resolving LLM caching failures and stabilizing agent performance.

Machine Learning Scientist Intern **Toronto, Ontario**
Fable Therapeutics *September 2025 - May 2026*

- Developed modular bioinformatics pipelines integrating BLAST+, ANARCI, and gemmi for multi-stage filtering, accelerating downstream candidate selection by 3x.
- Engineered a Hydra pipeline on a SLURM cluster, unifying isolated workflows to cut execution turnaround by 50%
- Automated dataset acquisition and preprocessing (10,000+ structures) using Dockerised downloaders and metadata-based extraction workflows.
- Leveraged AWS to run AlphaFold/abfold PDB remediation workflows, regenerating millions of missing atoms to eliminate discrepancies between experimental structures and recorded sequences.

Undergraduate Researcher in Computational Neuroscience **Toronto, Ontario**
Woodin lab *September 2024 - June 2025*

- Developed a CNN-based event detection tool that achieves a 94% accuracy and a 93% true positive rate, with the potential to reduce MINIS trace analysis time from weeks to minutes.
- Built a data preprocessing pipeline utilizing NumPy and Pandas to extract and label training data.
- Implemented data augmentation techniques to improve model generalization by synthesizing rare events from existing datasets.
- Continues to support ongoing MINIS research in the Woodin Lab through the CNN-based event detection tool developed with Melanie Woodin, President of the University of Toronto.

Software Engineer **Toronto, Ontario**
Factors Education *May 2024 - September 2024*

- Led a team of co-op interns in the development of serverless ingestion pipelines with Next.js, Node.js, and AWS (Lambda, S3, DynamoDB) to process and index video and text files into vector databases.
- Optimized vector storage by decoupling metadata from embeddings, reducing storage costs by ~75%.
- Enhanced the AI Video feature by integrating live transcript streaming via Deepgram, cutting transcript retrieval times by 30–50%.

- Implemented a Retrieval-Augmented Generation (RAG) pipeline with Pinecone and OpenAI embeddings to improve lesson quality for 6,000+ teachers, enabling context-aware teaching plans.

Undergraduate Researcher in Natural Language Processing **Toronto, Ontario**
Victoria University *January 2024 - May 2024*

- Cleaned and labeled datasets for the fine-tuning of LLMs such as ChatGPT 3.5 Turbo for enhanced model performance and accuracy in natural language understanding tasks.
- Streamlined literature reviews by architecting a custom GPT workflow to extract theories from qualitative papers into structured CSVs.
- Acquired a $2000 Vic Global scholarship to discuss the methodology and outcomes of my research at the Lisbon Scientonomy Workshop 2024.

Software Developer Intern **Toronto, Ontario**
Factors Education *September 2022 - September 2023*

- Co-Lead the construction of React, Flutter, and Streamlit (Python) product demos that directly contributed to over $75 000 in pre-seed stage funding.
- Implemented Google and Microsoft Federated user authentication mechanisms via AWS Cognito and hosted EC2 based web applications on AWS Route 53, ensuring high availability and scalability.
- Engineered a library of reusable, responsive React components to ensure UI/UX consistency across the production Data Dashboard, Landing Page, and core Platform.
- Collaborated with UI/UX designers and external developers from 321DataPro within an Agile (Scrum) framework to ensure seamless handoffs and standardized code documentation.

Computer Science and Math Tutor **Toronto, Ontario**
TeacherOn *May 2021 - June 2025*

- Orchestrated and managed personalized tutoring schedules for up to 10 hours a week for a diverse group of students throughout the academic year.
- Simplified and communicated algorithm design and Python programming fundamentals to students across university, high school, and primary education levels.
- Fostered a fun and engaging atmosphere, to help students find confidence in their learning abilities.

Student Researcher **Ottawa, Ontario**
Ashbury MSOE Center for Biomolecular Modeling SMART Team *September 2018 - May 2019*

- Published *The Effects of XIAP and Alzheimer's Disease* as part of the protein design and literature review teams.
- Worked alongside researchers and medical professionals at the Children's Hospital of Eastern Ontario (CHEO) to present their research.
- Engineered high-fidelity physical models of the XIAP protein by exporting molecular data from Jmol into AutoCAD for structural refinement and 3D printing preparation.
- Readied to present findings at the American Society for Biochemistry and Molecular Biology Conference (COVID canceled).

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, R, C, C++, Java

ML Frameworks and Tools: Claude, Cursor, PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas, Braintrust, AWS (Lambda, S3, DynamoDB, EC2, Route 53), Docker, Pinecone, GitHub

Full-Stack: Next.js, React, Node.js, FastAPI, PostgreSQL, Tailwind, REST